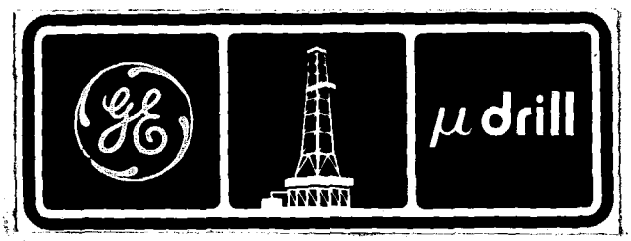
Instructions



μdrill™ 3000 Rig Operation

μdrill™ 3000
RIG OPERATION MANUAL

GEK-83762

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, U.S.A.

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CHAPTER 1
INTRODUCTION

This manual describes the location, purpose, and operation of the μdrill™ 3000 drive system controls and indicators. The μdrill™ 3000 is a modular rig drive system with individual SCR bridges for each motor. A typical rig drive system includes a driller's console, power control equipment (often mounted in a power control room), a mud pump console (optional), and the necessary motors and generators. Figure 1 is a simplified diagram of a typical μdrill™ 3000 rig drive system.

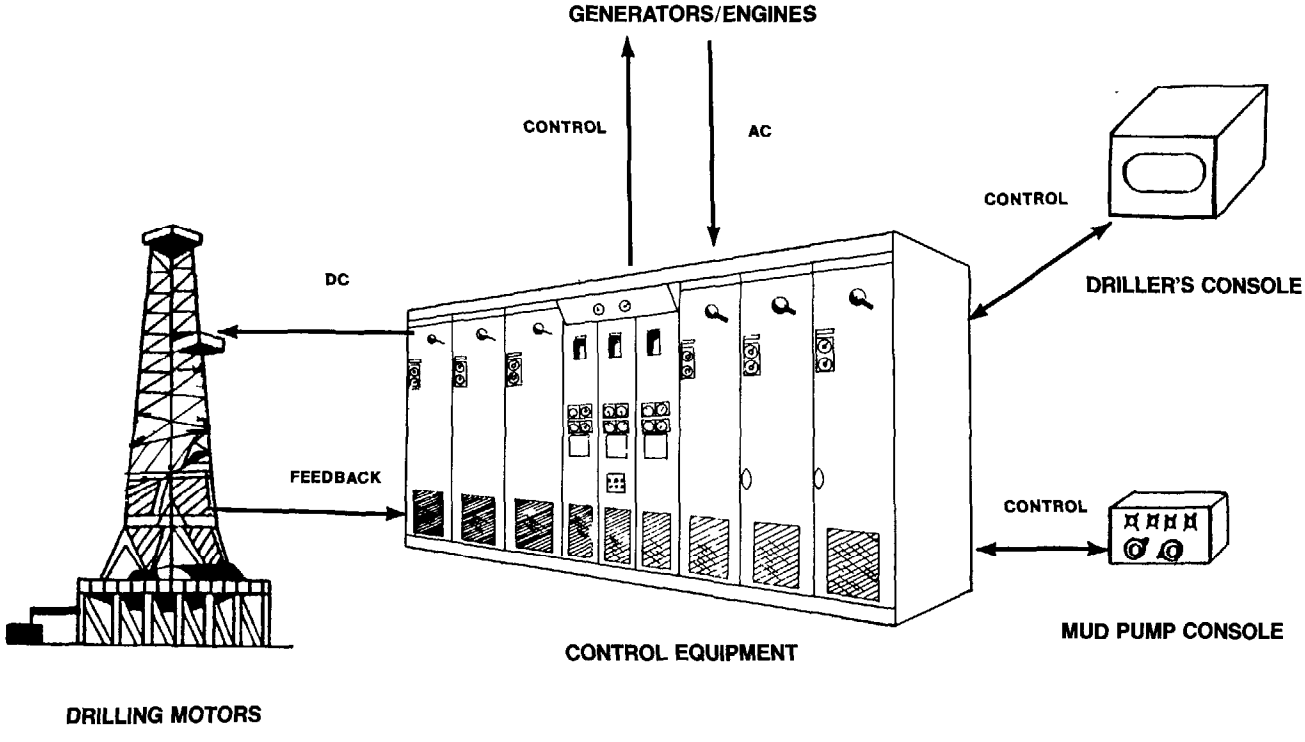


Figure 1.
Typical Rig Drive System

The **driller's console** contains controls such as selection switches (used to enable motor drives) and motor throttles. Indicators show equipment status such as motor load, SCR status, generator status, plus the status of oiler and blower motors.

The **mud pump console**, when used, contains a throttle and status lights for each mud pump. A switch on the driller's console determines whether the driller's console or mud pump console has control of the mud pump motors.

The **μdrill™ 3000 control equipment** contains a variety of controls and indicators, including generator and SCR controls, feeder breakers, ground detectors, and synchronizing equipment.

The μdrill™ 3000 rig drive system is available in a variety of configurations. There will be differences between rigs in such things as the number of motors, generators, and SCRs, the control functions available, whether the power control equipment is mounted in a power control room, and the types of options installed. Because of this, your rig drive system may not look exactly like some of the photographs or diagrams in this book. In addition, there may be descriptions of equipment or controls that your system does not have. Keep this in mind while using this manual, since all of the information may not apply to your particular rig.

CHAPTER 2
RIG CONTROLS AND INDICATORS

On the next several pages are photographs showing controls and indicators for the driller's console, mud pump console, and μdrill™ 3000 control equipment. Controls pointed out in the photograph are described in nearby text. See Chapter 3 for rig operating procedures.

DRILLER'S CONSOLE

The indicators, meters, throttles and switches on the driller's console give the driller control over the drilling motors. A description of each item on the driller's console is given below.

STATUS LIGHTS

These lights show the operating condition of rig equipment such as generators, SCRs, and motors. Each light is labelled with the name of the equipment or function monitored. When the light is on, the equipment or function is active. Some typical items that may be monitored by the status lights are listed below.

Gen On - The indicated generator is supplying power to the drive system when this light is on.

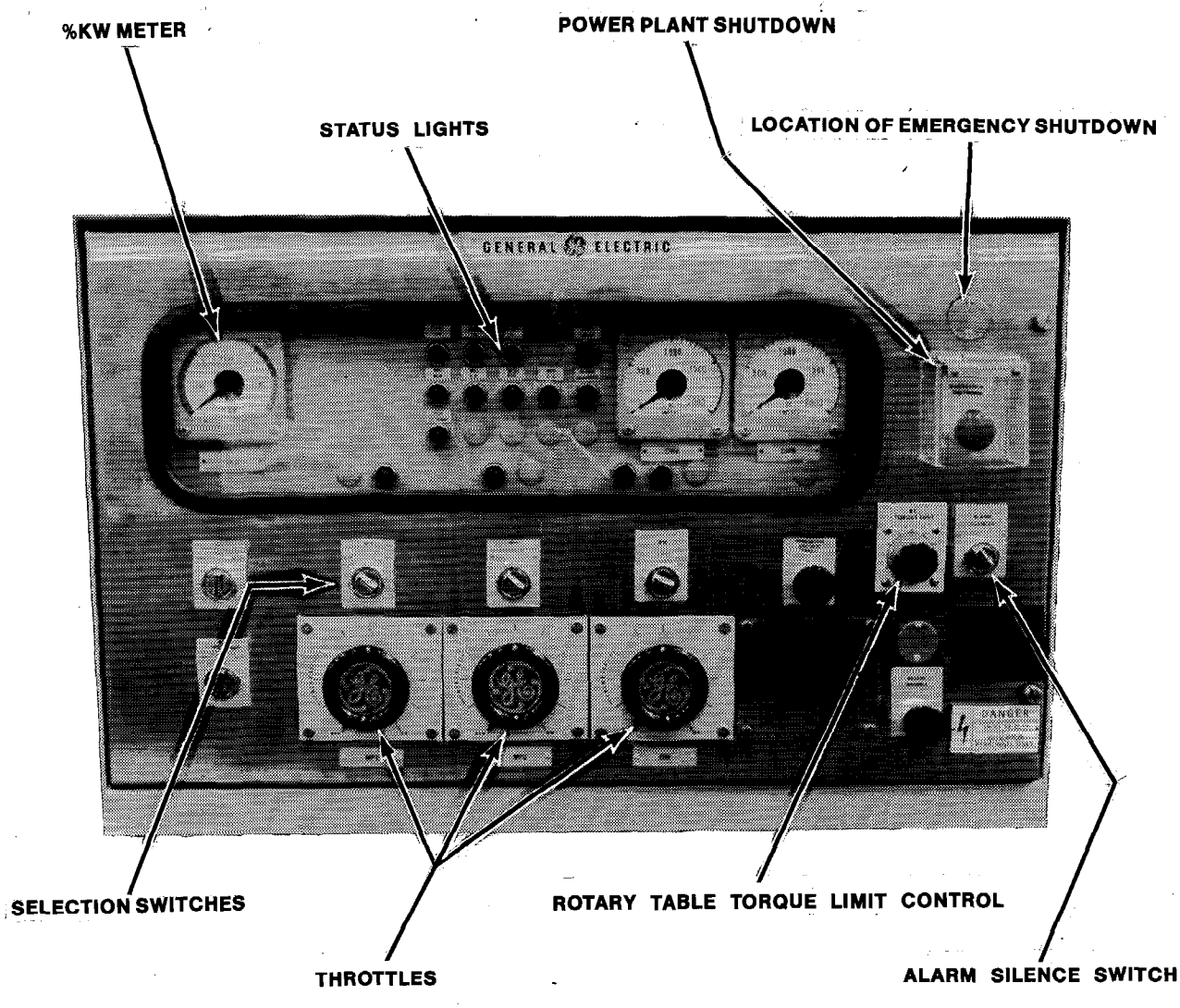
Drive Ready - The drive is selected and ready to run when this light is on. When the light flashes, it indicates that the drive is selected and awaiting permissives **or** indicates a warning if the drive is running.

Supercharger On - The supercharger is running when this light is on. When the light flashes, it indicates the supercharger failed to start.

Auxiliaries - The mudpump auxiliary motors are running when this light is on. When the light flashes, it indicates the auxiliary motors failed to start.

High Load - The engine/generator sets supplying power to the rig are approaching an overload condition when this light comes on.

Alarm Silenced - The alarm is silenced when this light is flashing. In this condition, the audible alarm will not sound for abnormal conditions.



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Figure 2.
Typical Driller's Console

SELECTION SWITCHES

These switches allow the driller to select specific drilling motor functions and directions. The number of selection switches depends on the number of drilling motor functions in the rig drive system.

Mud Pump Switches - A mud pump is selected by turning the desired switch (MP1 or MP2) to the ON position. Control of the mud pump is transferred to the mud pump console by turning the switch to the REM position.

Drawworks Switch - The drawworks are enabled when the DW switch is in either the FWD or REV position.

Rotary Table Switch - The rotary table is enabled when the RT switch is in either the FWD or REV position.

THROTTLES

The throttles control the speed of the motors indicated on the respective console nameplates. The number of throttles depends on the number of drilling functions in your particular configuration. A typical configuration would include four throttles: one for mud pump 1, one for mud pump 2, one for the drawworks, and one for the rotary table.

ROTARY TABLE TORQUE LIMIT CONTROL

This control limits the torque provided to the rotary table. The rotary table may be chain-driven by the drawworks motor or it may be an independent rotary with its own motor.

EMERGENCY SHUTDOWN

Pressing this button trips the SCR disconnects, thus removing motor field and armature power. ***Other power may still be connected to the motor at this time!***.

POWER PLANT SHUTDOWN

Pressing this button shuts off the field supply to the engine/generator sets, thus tripping the generator breakers. The button is mounted under a spring-loaded plastic cover to prevent accidental activation (optional). ***Complete shutdown of the engine/generator sets may require an air supply cut-off mechanism.***

%KW METER

This meter indicates how much of the total power available from the engines is being used by the rig. Using this meter, an operator can determine the correct number of engines to run to supply enough power for rig operation. The position of the needle indicates the number of engines required.

ROTARY TABLE TORQUE METER

This meter indicates the amount of torque applied to the drill pipe by the rotary table. The meter may be calibrated both in foot-pounds and amperes.

ALARM SILENCE SWITCH

When activated, this switch prevents the audible alarm from sounding at the driller's console. The **alarm silenced light** will flash as long as this switch is ON.

AUDIBLE ALARM

Unless disabled by the **alarm silence switch**, the **audible alarm** sounds in case of:

1. Loss of a motor blower while the motor is running.
2. Loss of the supercharger while a mud pump motor is running.
3. Loss of a mud pump auxiliary motor (i.e., a lube motor) while the mud pump motor is running.

MUD PUMP CONSOLE

The mud pump console is simply an extension of the driller's console and is used to control the mud pumps from a remote location.

THROTTLES

The mud pump throttles serve the same purpose on the mud pump console as on the driller's console: controlling the speed of the mud pump motors.

INDICATORS

There are normally four indicators on the mud pump console, one for each mud pump motor. These indicators are labelled DRIVE READY HERE and will light only if the following conditions are met:

1. The remote throttle must be selected by turning the mud pump selection switch on the driller's console to the REM position.
2. The auxiliary and supercharger motors must be operating.
3. The power converter must be operating.

CONTROL EQUIPMENT

The μdrill™ 3000 control equipment consists of one line-up, often located in a power control room. Figure 3 is a diagram of a typical control equipment line-up with major controls pointed out. This equipment controls the engine/generator, converts three-phase AC power to DC power for the drilling motors, synchronizes the power supplied by several generators, and detects ground fault conditions in the AC or DC power circuits.

Most of the controls on the line-up are used during rig start up and shut down. The following panels and modules contain controls and indicators that the operator should be familiar with. Each of these panels and modules is illustrated and described in this manual —

- Engine/Generator Control Panel
- Synchronizing Control
- Ground Detector Module
- Power Conversion Panel
- Feeder Breaker Panel
- Motor Control Center

ENGINE/GENERATOR CONTROL PANEL

The Engine/Generator Control Panel consists of indicators, meters, and switches used to start and control a generator and its prime mover. There is one Engine/Generator Control Panel for each generator. Figures 4 and 5 are photographs of a typical control panel showing the following controls and indicators:

Voltmeter - Indicates the AC voltage of a source selected by the **voltmeter switch**. The voltmeter can measure AC bus voltage or phase to phase voltage.

Ammeter - Indicates the current of the phase selected by the **ammeter** switch.

Kilowatt Meter - Indicates engine loading.

Kilovar Meter - Indicates generator loading.

Engine Control Switch - Sets the operating mode of the engine/generator engine. In the OFF position, the engine is shut down. In the IDLE position, the engine can be started and will operate at idle speed. In the RUN position, the engine operates at normal speed. In the LOAD position, the generator will load and the load sharing circuits of the generator are connected with the load sharing circuits of other on-line generators so that all generators will equally share the electrical load.

Voltmeter Switch - Selects the voltage source to be displayed by the **voltmeter**. Selecting VL1-2 causes the voltmeter to display the voltage between phases 1 and 2 of the generator output. Likewise, selecting VL2-3 causes the voltmeter to display the voltage between phases 2 and 3 and selecting VL1 causes the voltmeter to display the voltage between phases 3 and 1. Selecting VBUSS causes the VOLTMETER to display the voltage of the AC bus (phases 3-1).

Ammeter Switch - Selects the phase current to be displayed by the **ammeter**.

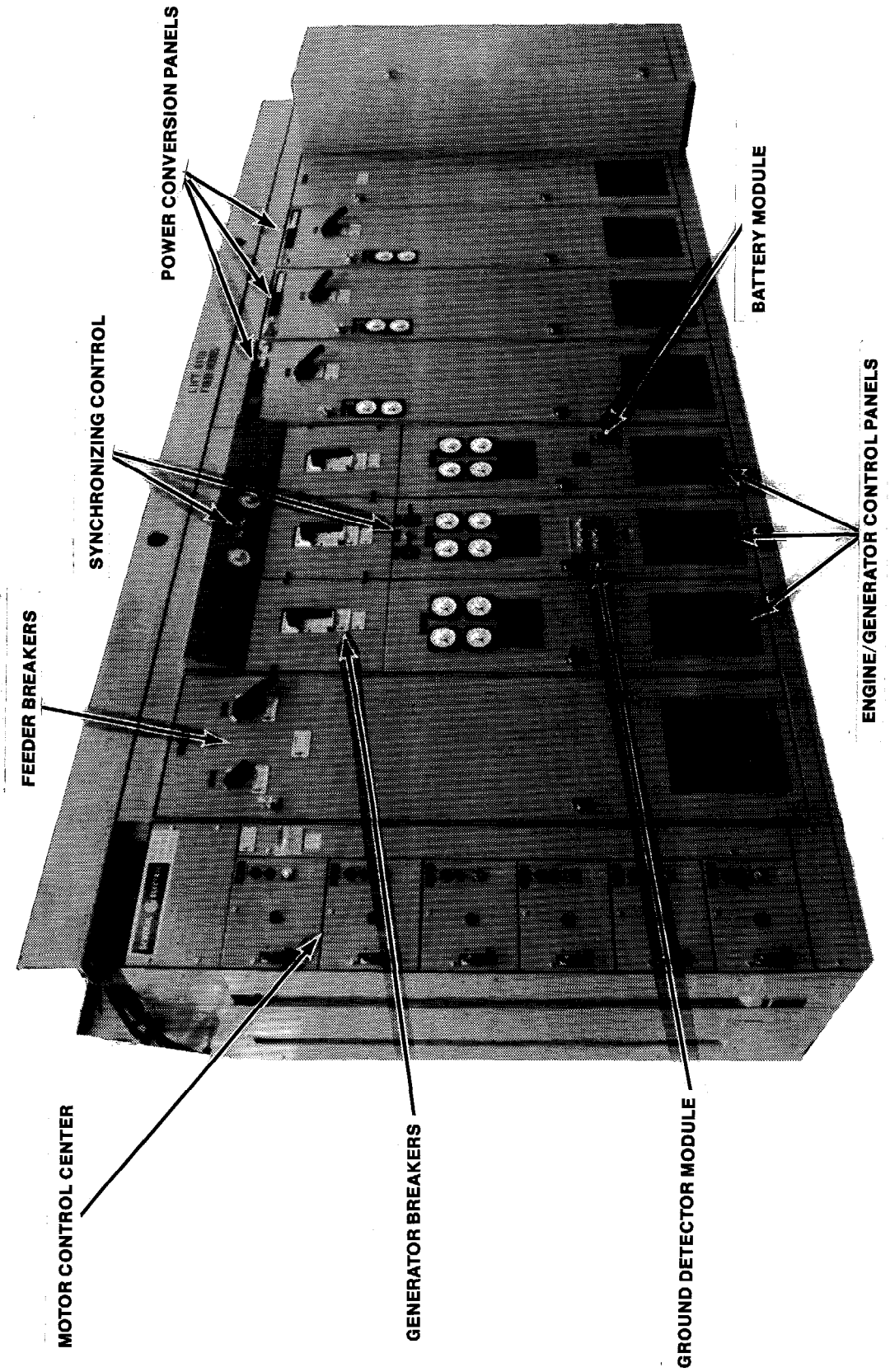
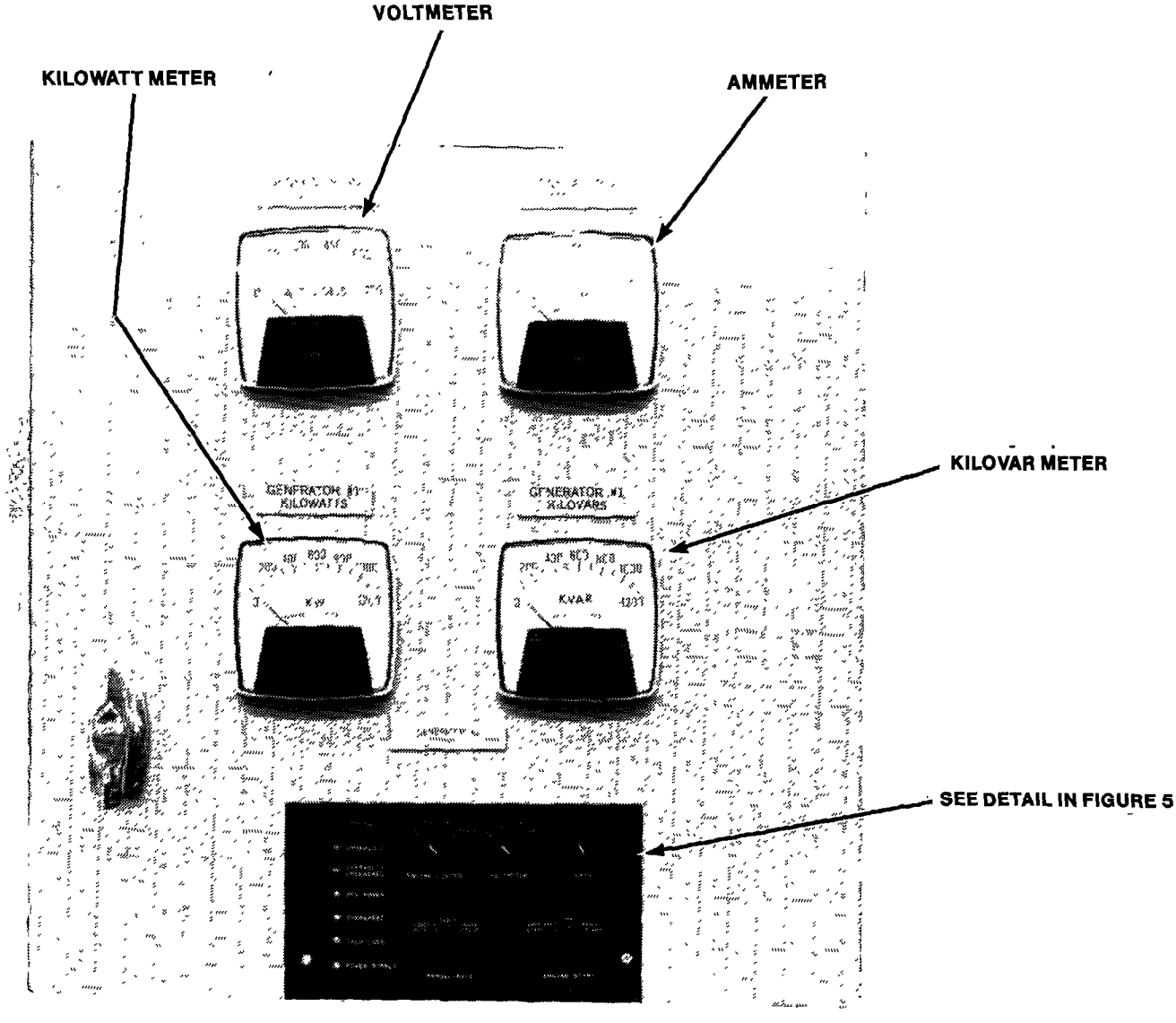


Figure 3.
Typical Control Equipment Line-Up

Manual-Auto Switch - Not used on this rig.

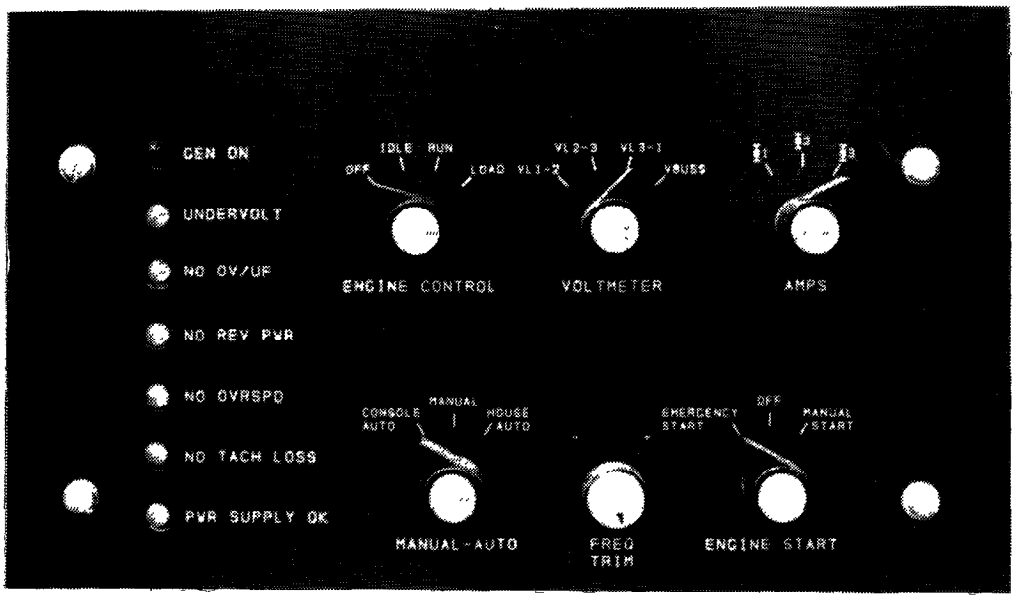
Frequency Trim Knob - Adjusts engine speed to control the generator voltage frequency.

Engine Start Switch - Not used on this rig.



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Figure 4.
Engine/Generator Control Panel



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Figure 5.
Engine/Generator Control Panel (Detail)

(NOTE: The following engine/generator control lights indicate whether the generator is operating normally or whether a fault has occurred. The first fault will cause its associated light to flash. Any subsequent fault will cause its associated light to go out.)

Gen On Light - When ON, indicates that the engine/generator is running, the breaker is closed, and power is being supplied to the AC bus. When OFF, indicates that the breaker is open.

Undervolt Light - When ON, indicates a generator voltage greater than 450 volts. When OFF, indicates that a generator undervoltage condition exists. This condition causes the **generator circuit breaker** to trip.

No OV/UF Light - When ON, indicates a normal condition. When OFF, indicates that a generator overvoltage or underfrequency condition exists. The **generator circuit breaker** will trip for an overvoltage condition, but not for an underfrequency condition.

No Rev Pwr Light - When ON, indicates a normal condition. When OFF, indicates that reverse power was flowing from the AC bus back to the generator. This condition causes the generator circuit breaker to trip.

No Ovrspd Light - When ON, indicates a normal condition. When OFF, indicates that the generator engine was overspeeding. Overspeeding causes fuel to be cut off from the engine and the generator breakers to open. ***The engine should be supplied with mechanical overspeed protection.***

No Tach Loss Light - When ON, indicates a normal condition. When OFF. indicates that the tachometer signal has been lost. This condition causes the generator engine to shut down and the generator breakers to open.

Pwr Supply OK Light - When ON, indicates a normal condition. When OFF, indicates that the generator control power supply has malfunctioned. This condition causes the generator engine to shut down and the generator breakers to open.

BATTERY MODULE

This module is located in the generator control section of the line-up. It is used to energize the controls of the first engine/generator module to be started up. The **battery module power switch** should always be in the ON position during rig operation so that the battery can be recharged. The switch must be turned OFF when the rig is shut down.

SYNCHRONIZING CONTROL

The controls and indicators of the synchronizing control are located in the generator control section of the line-up. The synchronizing control is used to synchronize the output frequency and phase of generators being brought on line with the frequency and phase of the 600 VAC bus. Failure to properly synchronize generators as they are brought on line can cause damage to the equipment. Figure 6 is a photograph of the synchronizing control equipment.

Frequency Switch - Selects the generator whose frequency is to be measured by the **frequency meter.**

Frequency Meter - Indicates the output frequency in Hertz of the generator selected by the **frequency switch.**

Synchronizing Switch - Selects the generator whose frequency is to be monitored by the **synchroscope** and connects the breaker closing circuit to the **breaker close pushbutton.**

Synchroscope - Indicates whether the output frequency of the generator selected by the **synchronizing switch** is faster or slower than the output frequency of the bus. Counterclockwise rotation of the needle means that the selected generator's output frequency is slower than the frequency of the bus. Clockwise rotation of the needle means that the selected generator's output frequency is faster than the frequency of the bus. A slow clockwise rotation is recommended for synchronizing.

Synchronizing Lights - Indicates the phase relationship between the generator selected by the **synchronizing switch** and the generator(s) already on line. The relative brightness of the lights indicates the degree that the generators are out of phase. When both lights are brightly lighted, the two generators are 180 degrees out of phase. When both lights are out, the two generators are in phase.

Breaker Close Button - When pressed, closes the circuit breaker of the generator selected by the **synchronizing switch** and brings the generator output on line to the 600 VAC bus. The generator's circuit breaker must be in the CHARGED position before the **breaker close pushbutton** will work.

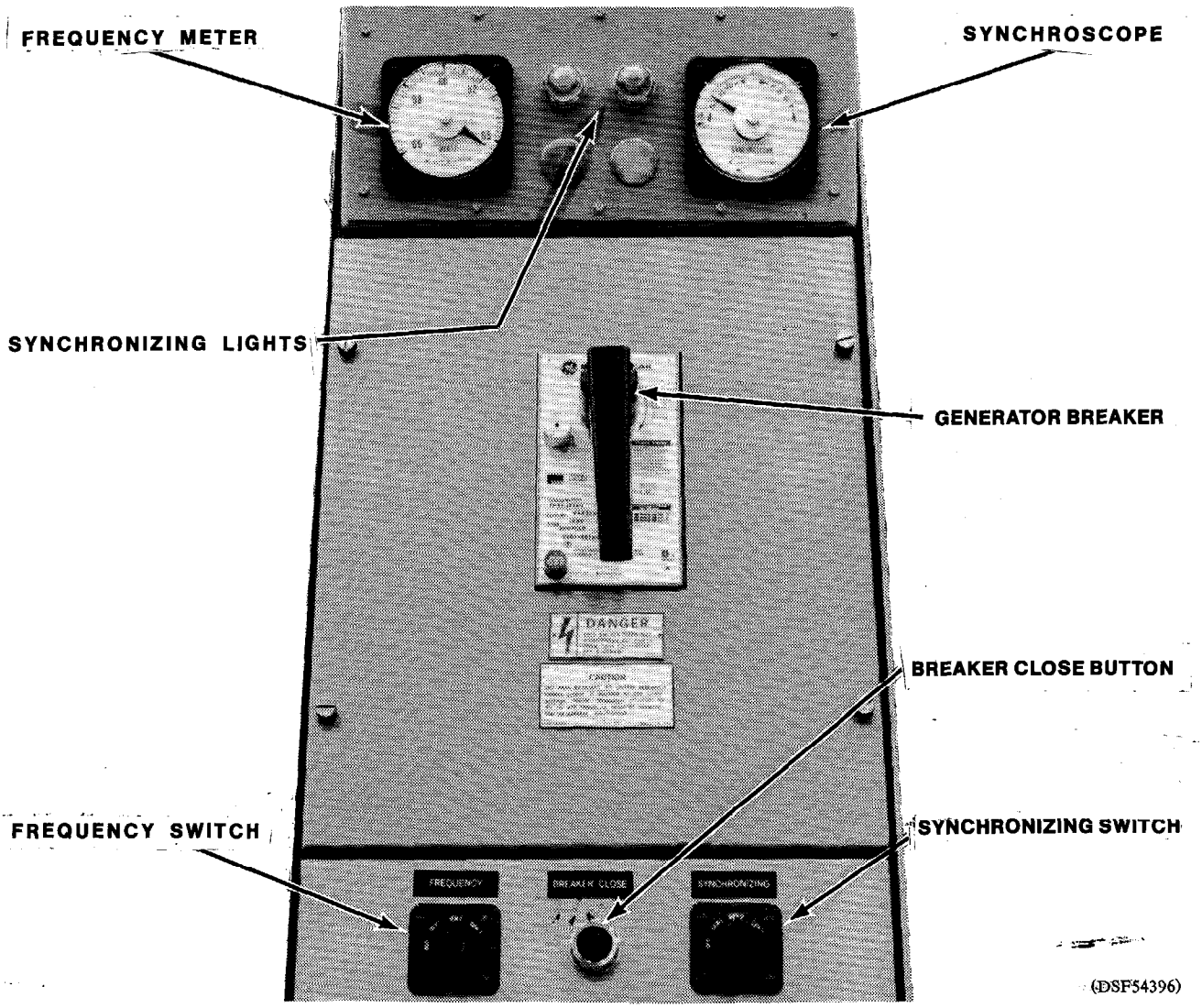


Figure 6.
Synchronizing Control

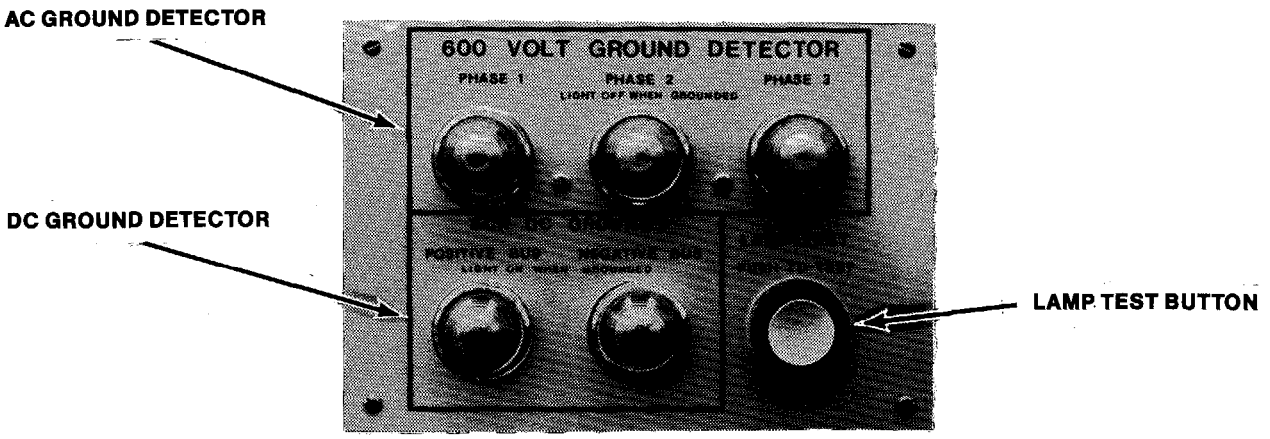
GROUND DETECTOR MODULE

This module, shown in Figure 7, detects ground faults within the AC and DC power system and displays the faults on the lights described below. A lamp test button allows the operator to check the lamps.

AC Ground Detector - These three lights correspond to the three phases present on the 600 volt AC bus. The lights are normally on; when a phase is grounded, the corresponding light turns off.

DC Ground Detector - These two lights correspond to the positive and negative SCR DC buses. The lights are normally off; when a bus is grounded, the corresponding light comes on.

Lamp Test Button - Pressing this button causes all AC ground detector lamps to light. This should be done periodically to detect faulty lamps.



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Figure 7.
Ground Detector Module

POWER CONVERSION PANEL

The power conversion panel uses SCRs (Silicon Controlled Rectifiers) to convert AC voltage supplied by the generators to a variable DC voltage for the drilling motors. As shown in Figure 8, the front of this panel contains two indicators, two meters, and a reset button that monitor the operation of the SCRs:

Voltmeter - This meter indicates the DC voltage supplied to the drilling motor by the SCR.

Ammeter - This meter indicates the DC current supplied to the drilling motor by the SCR.

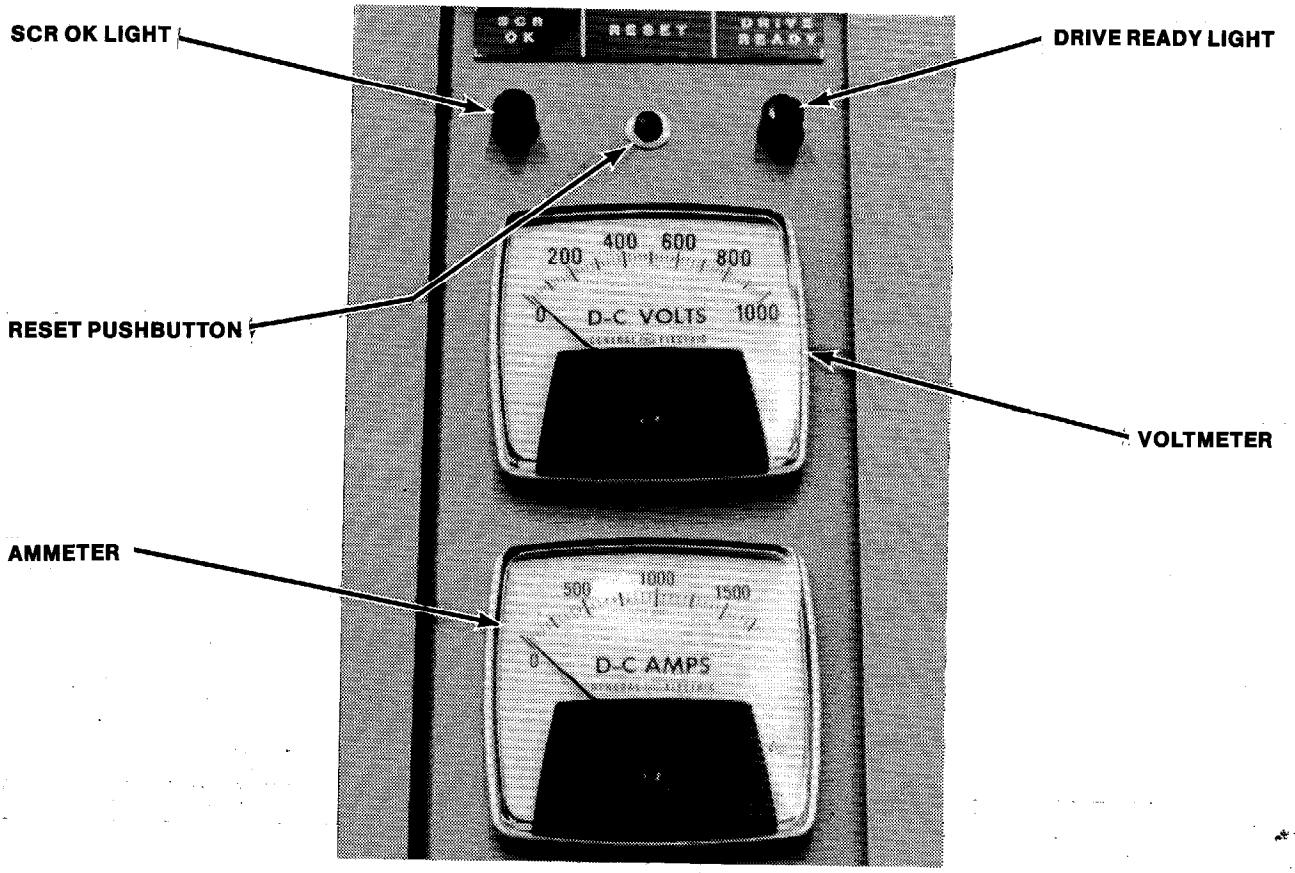
SCR OK Light - When ON, indicates that the SCR is available for drive selection.

Drive Ready Light - When on, indicates that the drive is ready to run. When off, indicates that the drive is not selected. When flashing, indicates that the selected drive is awaiting permissives **or** indicates a warning if the drive is running.

Reset Pushbutton - Pressing this button causes the power conversion panel to check itself for shorted cells or failure of a cell to fire. This button is active only when the drive is **not** selected.

FEEDERS

The **motor control center circuit breaker** connects and disconnects power to the **motor control center**, which is a collection of starters for auxiliary rig motors such as blowers and lube motors. The **lighting transformer circuit breaker** connects and disconnects power to the lighting transformer, which supplies power for lights and convenience outlets on the rig.



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Figure 8.
Power Conversion Panel Indicators

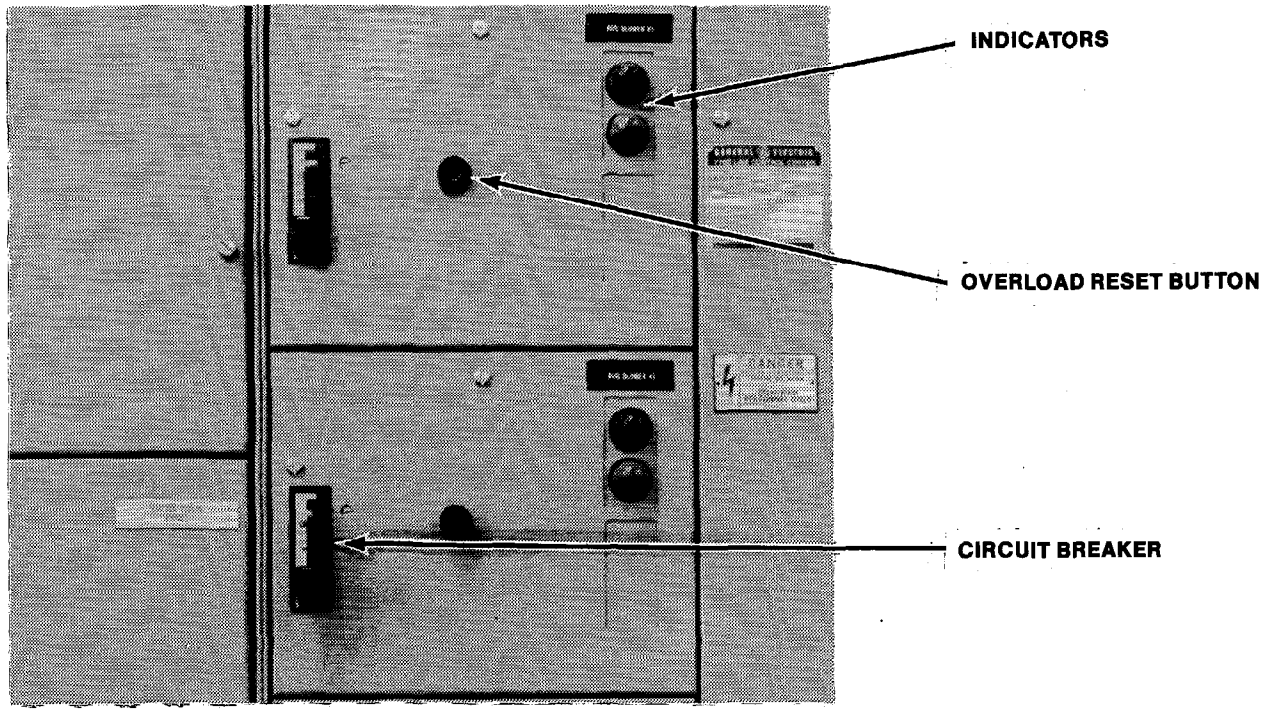
MOTOR CONTROL CENTER

The Motor Control Center (if supplied with the lineup) contains motor starters and feeder breakers for various rig electrical functions (lighting, auxiliary motors, motor blowers, etc.). The motor control center can be either 600VAC or 480 VAC. Figure 9 is a diagram of a portion of a typical Motor Control Center showing the following controls and indicators:

Circuit Breaker - This is a three position switch that connects or disconnects power to the starter. In the ON position, power is applied to the starter. If the circuit breaker is in the TRIP position, AC power has been removed from the starter because of an overcurrent or automatic trip condition. To reset the circuit breaker, move the switch to the OFF position and then back to the ON position.

Overload Reset Button - This button resets a starter that opened because of an overload. Since an overload will not trip the circuit breaker, if the motor stops and the circuit breaker has not tripped, wait five minutes and then press the overload reset button to reset the starter. The five minute time delay is required to allow the thermal overload trip device to cool.

Indicators - The red and green (optional) indicators show whether or not the breaker is closed and the motor starter is energized. If the green light is on, the breaker is closed, but the motor is not running. If the red light is on, the starter is energized and the motor should be running. If neither light is on, the breaker is open.



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Figure 9.
Portion of Typical Motor Control Center

NOTES

CHAPTER 3
OPERATING PROCEDURES

The procedures outlined below are for normal, day-to-day operation of the μdrill™ 3000 rig drive system, including rig start up, operation, and shut down.

These procedures assume a rig drive system with typical options installed. Your system may not have some of the controls or indicators mentioned. In addition, rig drive systems vary in the number of generators, generator controls, power control panels, and motors. Disregard any steps that do not apply to your rig drive system.

In each of the steps below, the names of modules, panels, indicators, and controls which are described in this manual are printed in **bold type**. For descriptions of other devices, refer to the appropriate vendor publications.

RIG START UP

This procedure describes how to start up a typical μdrill™ 3000 rig drive system, assuming that the rig, drive system, generators, and engines have been properly installed and checked out.

1. Preliminary steps:
    a. Ensure that the **power plant shutdown** and **emergency stop** switches on the **driller's console** are pulled out.
    b. Deselect all motors and turn all throttles off at the driller's console.
    c. Position the **battery module power switch** to the ON position.
    d. Ensure that the **engine control switch** for the generator being started is in the IDLE position.

2. Start the desired engine and follow the manufacturer's recommended warm up procedures before continuing with the following steps. (The following steps assume that generator 1 is being brought on line first, followed by generator 2.)

3. Set the **synchronizing switch** to GEN 1.

4. Set the generator 1 **engine control switch** to RUN. All of the indicating lights except GEN ON on the generator 1 **engine/generator control panel** should come ON. Check that the **generator voltmeter** for generator 1 reads 600 volts.

5. Set the **frequency switch** on the **synchronizing control** to GEN 1. Turn the **frequency trim knob** until the **frequency meter** reads 60 Hertz.

6. Set the generator 1 **engine control switch** to LOAD.

7. Charge the **generator breaker** and press the **breaker close switch** to bring the generator on line to the AC thru bus. The GEN ON light should come on.

8. Close the **lighting transformer circuit breaker** and the **motor control center circuit breaker.** Start the essential motors (i.e., air compressors, fuel pumps, radiator fans, etc.) from the **motor control center.**

9. Start the generator 2 engine according to the instructions in steps 1 through 6 above. Check the **generator voltmeter** for generator 2 and make sure it reads 600 volts.

10. You will now use the **synchroscope** to synchronize the output frequency of the two generators before you bring generator 2 on line. The **synchroscope** indicates whether the output frequency of the generator selected by the **synchronizing switch** is higher or lower than the output frequency of the generator already on line. Set the **synchronizing switch** on the **synchronizing panel** to GEN 2. Adjust the **frequency trim knob** on the generator 2 **engine/generator control panel** until the **synchroscope** needle is rotating slowly in a clockwise direction (about one revolution every 15 seconds).

11. Charge the **main generator breaker** for generator 2.

12. Watch the **synchroscope** and press the **breaker close switch** just before the rotating needle reaches the twelve o'clock position. This brings generator 2 on line to the AC bus. The GEN ON light should come on.

13. The meters on the generator 2 **engine/generator control panel** will now indicate Generator 2 voltage, current, and power. The values on these meters should be the same for all other on-line generators.

14. If your rig uses three or more generators, repeat steps 9 through 13 for the additional engine/generator sets.

15. Check that the **AC disconnect switch** on the front of each **power control panel** is closed to provide AC power to the SCRs and control logic. At this point the SCR OK light should be ON and the SCRs ready to be selected.

16. Energize the appropriate auxiliary motor starters from the **motor control center.** These would include drill motor blowers, mud pump oiler motors, and so forth. These motors will start automatically when the drilling motor assignments are made from the **driller's console.** ***Drilling motor cut-out switches must be un-pinned before the drive will operate.***

The rig drive system is now ready for operation.

RIG OPERATION

Once the rig drive system has been started, the drilling motors are controlled from the driller's console and mud pump console. The operator will seldom need to concern himself with the control equipment.

Most operations can be broken down into four types of activities:

- Making motor selections (including alternate assignments)
- Controlling motor speed and direction
- Monitoring rig operation
- Correcting operating problems

These activities are described below. As we describe each activity, refer back to the photograph of the driller's console (Figure 2).

MAKING MOTOR SELECTIONS

The **selection switches** on the driller's console are used to enable SCR power converters for various drilling functions and directons. Each **selection switch** represents a specific drilling function; for example, mud pump #1, the drawworks, or the rotary table.

Once the motor has been selected, the DRIVE READY light for that motor will come on and the motor can be controlled with the **throttle**. If the rig has a purge option, the DRIVE READY light will flash for six seconds before coming on. If the light flashes for more than six seconds, one or more of the following problems exist:

- The motor blower may have failed to start or the signal from it may have failed to reach the **power conversion panel**.
- The **throttle** at the **driller's console** may not be in the OFF position.
- The field or reversing contactor may have failed.
- An auxiliary drive motor (optional) may have failed to start.

CAUTION

If the DRIVE READY light begins to flash during drilling, a problem has arisen. Continued operation under these conditions may cause damage to equipment.

Troubleshooting of this equipment is covered in the ***μdrill™ 3000 Troubleshooting Guide.***

Alternate Assignment (Optional)

If your rig contains the alternate assignment feature, there will be an alternate assignment selection switch on the driller's console. The alternate assignment option allows the driller to use a single power conversion panel for more than one drilling function. The most common alternate assignment is to use the drawworks "B" power conversion panel to power the drawworks "B" motor (normal) or the rotary table motor (alternate assignment). Other alternate assignments are available but all typically operate the same.

Using the drawworks "B" alternate assignment option described above as an example, a typical alternate assignment operation is as follows. In the normal mode, the drawworks "B" power conversion panel will supply power to the drawworks "B" motor. In the alternate assignment mode, the drawworks "B" power conversion panel will supply power to the rotary table and the drawworks "B" motor will be disabled. The mode of operation is selected using the alternate assignment selection switch on the driller's console. After selection of alternate assignment all controls, indicators, and auxiliaries will operate normally for the alternate motor.

CONTROLLING MOTOR SPEED AND DIRECTION

The **driller's console** is the main control center for all rig motors. The driller can control both motor speed and direction with the motor throttles and selection switches.

Mud Pump

The driller selects a mud pump by turning the proper **selection switch** to the ON position. The pump is then controlled using the corresponding **throttle**. If the rig has an optional **mud pump console**, the driller can transfer control of the mud pump to that console by turning the **selection switch** to the REM position. With the **selection switch** in the REM position, the DRIVE READY lights on the **mud pump console** will light. If these lights begin flashing, there is a problem (as described above under "Making Motor Selections") that should be corrected. If the lights illuminate continuously, the mud pumps are ready to run.

Under normal operating conditions, two motors on the mud pump share the pumping load. If one-motor operation is required, the motor not in use should be de-assigned at the power control room. When this is done, the μdrill™ 3000 drive system will automatically recalibrate the current limit on the remaining motor, allowing that motor to supply more power than it would under two-motor operation.

Drawworks

The drawworks are controlled from the **driller's console** only. The driller controls the drawworks using a single **selection switch** and a single **throttle**. The **selection switch** enables operation of the motor in either the forward (FWD) or reverse (REV) direction. The FWD position of the switch enables regenerative braking of the drawworks motors. Regenerative braking is a method of slowing down a motor by pumping its energy back into the AC power system, thus allowing the motor to decelerate more rapidly.

Drawworks motor speed is controlled by either a hand or foot **throttle** at the **driller's console**. Usually, cathead speed is set with the hand throttle and hoist speed is controlled with the foot throttle. Deselection of the drawworks motor must be done in the power control room.

Rotary Table

The rotary table is driven by its own motor, which is controlled by a **selection switch** and **throttle** on the **driller's console**. The **rotary table torque limit control** is used to limit the torque applied to the drill string. This torque limit also limits cathead torque. The μdrill™ 3000 drive system is the first rig system to allow the driller to control cathead torque.

The rotary table may be chain-driven and clutched off the drawworks motors. In this case, the driller uses the **drawwork selection switch** and **throttle** to control the rotary table. Clutching determines the correct current limit value for drawworks or rotary table operation. Whenever the drawworks is unclutched, the rotary table torque limit controls cathead and rotary torque.

MONITORING RIG OPERATION

There are several indicators and meters on the consoles that should be monitored during rig operation. Make a habit of occasionally scanning the console indicators and meters to ensure proper rig operation and early detection of problems.

Should any of the **status lights** show an abnormal condition the **audible alarm** will sound, unless the **alarm silence switch** is ON. Leave the silence switch OFF unless you need to silence an active alarm while you fix a problem.

Before starting a drilling operation or making a function change, the driller should check the following things:

1. Are the correct number of GEN ON lights ON?
2. Are all **selection switches** set properly and are the correct DRIVE READY lights on?
3. Are all **throttles** in the OFF position?
4. Is the **rotary table torque limit control** set according to the drill pipe grade and diameter?
4. Is the drawworks **selection switch** set in the correct direction (FOR or REV)?

CORRECTING OPERATING PROBLEMS

Two common operating problems are described below along with suggestions for correcting them:

Insufficient Rig Power - If the speed of all DC motors decreases and the **KVA/KW limit** light comes on, there are not enough engines/generators on line to support the heavy drilling load. Should this occur, the load must be reduced or an additional generator brought on line.

Insufficient RT Torque - If only the rotary table's speed decreases, the torque limit set by the **rotary table torque limit control** has been reached. Ensure that the limit control is set properly for the type of pipe being used. If the setting is correct, you must reduce the torque requirements. There may be too much weight on the bit or the wrong gear may have been selected.

RIG SHUT DOWN

This procedure describes how to completely shut down the μdrill™ 3000 rig drive system in preparation for power off maintenance or moving of the rig.

1. Turn all **throttles** on the **driller's console** to the OFF position. Turn all **selection switches** to the OFF position.
2. Stop all non-essential loads.
3. Turn the **engine control switch** on the **engine/generator control panel** to the RUN position. This unloads the engine. Turn the **generator circuit breaker** to the OFF position. Repeat this step for each generator.
4. Turn the the **engine control switch** on each **engine/generator control panel** to the IDLE position. After the manufacturer's recommended idle time, turn the **engine control switch** to the OFF position.
5. Turn the **battery module power switch** to the OFF position.